BOUBAKAR DIALLO

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EDUCATION

Cornell University **B.S. Electrical & Computer Engineering** **May 2028**

Honors Community Leadership Award - Cornell University (Aug 2025)

Coursework Computer Architecture, Digital Logic & Computer Organization, Embedded Systems, Computer Systems Programming, Signals & Systems, Circuits, Linear Algebra, Foundations of Robotics

SKILLS

RTL & Microarchitecture Verilog/SystemVerilog, RISC-V, Processor Datapath & Control, Five-Stage Pipelining Data Forwarding, Hazard Detection, Stall/Squash Logic, FSMs, Iterative Multipliers

Verification & FPGA Design RTL Simulation, Directed & Randomized Testing, pytest Regression Testing, Val/Rdy Protocols, VCD Waveform Debugging, Vivado, Quartus, FPGA Synthesis, Timing Analysis

Programming & Tools Python, C/C++, ARM Assembly, Bash, Linux, Git/GitHub, Oscilloscopes, Logic Analyzers

PROJECTS

Pipelined RISC-V Processor & Variable-Latency Multiplier **Sep 2026**

- Developing a five-stage TinyRV2 RISC-V processor with stalling and fully bypassed pipelines, implementing X/M/W forwarding, RAW hazard detection, and stall/squash control.
- Designed fixed- and variable-latency iterative multipliers with FSM-based control and zero-bit skipping, integrating a 32-bit multiplier via val/rdy handshakes and pipeline stall coordination.
- Verifying processor RTL against Python reference models using directed and randomized pytest regressions with injected memory delays, debugging hazards through execution traces and VCD waveforms.

RISC-V Processor & Custom Hardware Accelerator **Dec 2025**

- Implemented and FPGA-prototyped a single-cycle TinyRV1 processor with a 32-bit datapath, register file, instruction decode, ALU control, and memory-mapped peripherals; validated using RTL simulation and hardware I/O testing.
- Designed a custom 32-bit accumulation accelerator with FSM control, a 16-bit counter, and dedicated datapaths, achieving 4.09√ó speedup (131 to 32 cycles) over TinyRV1 on a 31-element workload.

Preemptive RTOS & Embedded Peripheral Drivers **May 2026**

- Designed gate-level and RTL implementations of a 16-bit counter, reducing critical-path delay from 24.553 ns to 7.774 ns (68.3%) and logic utilization from 1,545 to 109 ALUTs.
- Implemented a Moore FSM-based FPGA music player with counters, registers, and note-generation logic, validating functionality through simulation and oscilloscope measurements.

Preemptive RTOS & Real-Time Scheduling **Jul 2026**

- Developed a preemptive RTOS on ARM Cortex-M0+ with context switching, process-state queues, spin/blocking locks, and Earliest Deadline First scheduling using 1 ms timer interrupts.
- Implemented deadline-ordered ready queues, delayed task activation, and preemption; validated scheduling and synchronization through concurrent-process tests covering lock contention, race conditions, and deadlines.

WORK EXPERIENCE

Software Engineer Intern *Simulacrum, NYC, NY* **Jan – Jun 2026**

- Engineered a distributed real-time sensor-data platform using FastAPI and Google Cloud Pub/Sub, processing high-throughput time-series streams with sub-100 ms latency for forecasting and model-training workflows.
- Built observability pipelines with structured logging and automated alerts for latency spikes, data drift, and processing failures, reducing debugging cycles by approximately 30%.

Embedded Systems Intern *ADAS SAFE, Jacksonville, FL* **Jun – Aug 2025**

- Diagnosed microsecond-level synchronization drift across radar, IMU, and LiDAR hardware, identifying timing dependencies that informed sensor-fusion hardware/software interface design.
- Validated automotive perception hardware, isolating calibration and synchronization faults and reducing radar alignment errors by 15% across production deployments.

Robotics & Hardware Systems Lead *Engineers Without Borders, Ithaca, NY* **Oct 2024 – Jan 2026**

- Designed and fabricated custom PCBs, motor-driver circuits, and power-regulation subsystems for an agricultural rover based on NASA's open-source Mars rover architecture.
- Led board bring-up and hardware/software integration, diagnosing power and sensor-interface faults using logic analyzers and integrating a ROS-controlled robotic arm.

Incoming Technology Intern *PNC Bank* **Summer 2027**